Road Damage Detection and Risk-Aware Geospatial Routing for Road Maintenance Prioritisation

Bachelor's Thesis

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Abstract: Road surface damage raises driving risk and maintenance costs, yet manual inspection does not scale to municipal road networks. This thesis presents and evaluates an end-to-end pipeline that detects road damage in street photographs, converts detections into interpretable risk scores, and uses those scores for risk-aware routing and maintenance prioritisation. YOLOv8n and Faster R-CNN were trained under matched conditions on a class- and country-balanced subset of RDD2022. YOLOv8n achieved higher mAP50-95 (0.300 versus 0.186) and substantially faster inference (median 22 ms versus 307 ms per image). Both models dropped sharply on manually collected photographs from Sofia (mAP50-95 of 0.041 and 0.040 respectively), and fine-tuning on 131 local images raised YOLOv8n's F1 from 0.19 to 0.35. Against independently labelled Sofia images, the rule-based risk classifier reached 59.5% accuracy (macro F1 0.58, below the 0.75 target); its errors traced to missed detections rather than mis-scored ones. In a neighbourhood with recorded hazards, risk-aware routes reduced hazard exposure by a median of 37% (mean 46%) compared with shortest paths, with 95% of routes within a 10% distance increase (Wilcoxon $p = 0.0017$). Generated segment risk correlated with manual ratings (Spearman's $\rho = 0.68$). A lightweight detector adapted with modest local data, combined with explainable risk scoring, can support municipal maintenance decisions, with detection recall and local data coverage as the main constraints.

1 Introduction

1.1 Context and problem

Road surface damage such as potholes, longitudinal cracking, transverse cracking, and alligator cracking degrades driving safety and comfort, and represents a recurring cost for municipal road authorities. Damage that is left unrepaired tends to worsen over time, so identifying it early and prioritising repairs accordingly is central to cost-effective road maintenance. The conventional approach, manual visual inspection by trained personnel, does not scale well to an entire municipal road network: it is labour-intensive, infrequent by necessity, and produces assessments that are difficult to compare consistently across inspectors or over time.

Computer vision offers a scalable alternative. Convolutional object detectors can be trained to localise and classify road damage directly from photographs, removing the dependency on continuous manual inspection and enabling damage records to be produced automatically from images collected by ordinary vehicles, phones, or dedicated inspection rigs. A large body of work, reviewed in Section 2, has demonstrated that this is technically feasible for the detection step in isolation.

Detection alone, however, does not constitute a decision-support system. A list of bounding boxes and class labels does not, by itself, tell a municipality which streets to repair first, nor does it change how a route is planned through a network that already contains known hazards. Turning detections into something actionable requires two further steps: translating each detection into an interpretable measure of risk, and propagating that risk into the geospatial systems, such as road networks and routing algorithms, that municipalities and drivers already use. This thesis treats detection, risk scoring, and risk-aware routing as one connected pipeline rather than as three separate problems, on the argument that the practical value of automated road-damage detection is realised only once its output is connected to a decision such as "repair this segment first" or "avoid this street".

1.2 Research questions and hypotheses

The central research question addressed in this thesis is:

> *How can computer vision-based road damage detection be integrated with risk-based geospatial routing to improve route safety and support road maintenance prioritisation?*

This question is decomposed into four sub-questions, each associated with one hypothesis that is tested empirically in Section 4.

Sub-question 1. Which object detection model performs best, based on evaluation metrics such as mean average precision (mAP), precision, and recall, for detecting road damage in images?

> **H1:** YOLOv8 will achieve detection accuracy (mAP) within 2 percentage points

(mAP50-95) of Faster R-CNN, while maintaining lower computational complexity. A 2% difference in mAP is considered an acceptable performance margin for comparing models of broadly similar accuracy.

Sub-question 2. How can detected road damage be translated into meaningful road risk scores for routing and maintenance prioritisation?

> **H2:** A rule-based risk scoring model based on damage type and size will effectively differentiate road segments into low, medium, and high-risk categories.

Sub-question 3. How does incorporating road damage risk into routing algorithms influence route selection compared to traditional shortest-path routing?

> **H3:** Risk-aware routing will reduce route hazard exposure compared to shortest-path routing, while maintaining a reasonable increase in total route distance.

Sub-question 4. To what extent can the proposed system support municipalities in identifying and prioritising road maintenance more efficiently?

> **H4:** The system will enable identification of high-risk road segments, which can support maintenance prioritisation decisions based on aggregated risk scores.

1.3 Approach overview

The detection stage of the pipeline is trained on RDD2022 (Arya et al., 2024), a multi-national road damage image dataset, using two object detection architectures evaluated under matched conditions: YOLOv8n (Jocher et al., 2023), a single-stage detector, and Faster R-CNN with a ResNet50-FPN backbone (Ren et al., 2015), a two-stage detector. Both architectures are trained on three dataset scales and, at the final scale, on an explicitly class- and country-balanced subset of RDD2022, so that the comparison in Section 4.1.3 isolates the effect of architecture rather than an accidental difference in training data. Detected damage is converted into a per-detection risk score using a fixed, explainable rule based on damage class and detection confidence, and combined into image- and road-segment-level risk using the same combination rule at both levels. Road segments are identified using OpenStreetMap data accessed via OSMnx (Boeing, 2017), and segment-level risk is used to bias a shortest-path routing algorithm away from known hazards. Because a model trained only on an international benchmark dataset is not guaranteed to generalise to a specific deployment area, the pipeline additionally includes a domain-adaptation stage: both detectors are fine-tuned on a small set of manually collected photographs from a real pilot neighbourhood (Sveta Troitsa, Sofia, Bulgaria), and the resulting system, from detection through to routing, is evaluated against real, independently collected ground truth from that same area. Figure 1 summarises the pipeline and the hypothesis tested at each stage.

1.4 Contribution

This thesis makes three contributions. First, it provides a model-comparison between YOLOv8n and Faster R-CNN that is repeated at three dataset scales under matched training conditions, and tested with formal statistical procedures (a paired significance test and a bootstrap confidence interval, rather than a single point estimate) rather than being asserted from one training run. Second, it quantifies the generalisation gap between a detector trained purely on an international benchmark and its performance on real, previously unseen photographs from an actual deployment area, and measures how much of that gap a small, low-cost domain-adaptation step can close. Third, it tests all four hypotheses above against real, independently collected ground truth rather than treating any of them as self-evidently true: a manually labelled risk-classification sample for H2, a systematically generated route sample for H3, and manually rated road segments for H4, each analysed with an appropriate statistical test.

2 Related Work

2.1 Road damage detection with computer vision

Automated road damage detection has moved from small, single-country image collections toward large, multi-national benchmarks. RDD2022 (Arya et al., 2024) is one such benchmark, aggregating road images from seven countries (Japan, India, the Czech Republic, Norway, the United States, and two Chinese collection methods) and standardising damage annotation around a small set of classes: longitudinal cracking, transverse cracking, alligator cracking, and potholes, the same four classes used throughout this thesis. Prior work built on this and comparable benchmarks has repeatedly demonstrated that convolutional object detectors, and single-shot detectors of the YOLO family in particular (Redmon et al., 2016; Jocher et al., 2023), can localise and classify these damage types with useful accuracy directly from photographs. The Crowdsensing-based Road Damage Detection Challenge (Arya et al., 2022a), run on this same dataset, reported ensemble submissions

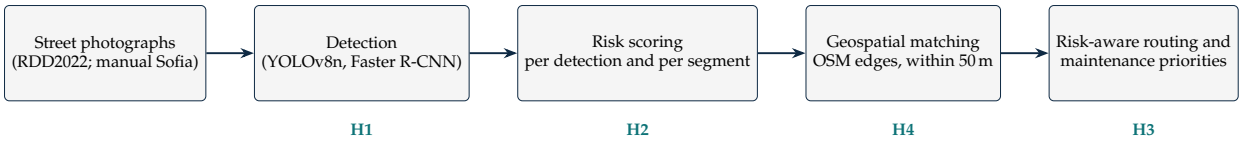


Figure 1: Pipeline overview. Detections are converted into risk scores, matched to OpenStreetMap road segments, and used for risk-aware routing and maintenance prioritisation. H1–H4 mark the hypothesis tested at each stage.

reaching an F1 score of around 0.76 on the held-out challenge test set, giving a concrete, independently verified benchmark figure against which the detector accuracies obtained in this thesis (Section 4.1.3) can be contextualised, bearing in mind that the CRDDC figure comes from multi-model ensembles rather than the single architectures compared here. This thesis reuses RDD2022 directly as its primary training source (Section 3.1), rather than proposing a new dataset, and instead focuses on a rigorous comparison of detector architectures and on what happens once such a detector is deployed on real, previously unseen images from one specific city.

2.2 One-stage vs. two-stage detectors

Object detectors are broadly divided into two-stage detectors, which first propose candidate regions and then classify and refine each one, and one-stage (single-shot) detectors, which predict boxes and classes directly over a dense grid in a single forward pass. Faster R-CNN (Ren et al., 2015) is the canonical two-stage architecture, combining a region proposal network with a downstream classification and bounding-box regression head; the YOLO family (Redmon et al., 2016; Jocher et al., 2023) is the canonical one-stage alternative. The conventional framing in the literature is a speed-versus-accuracy trade-off: two-stage detectors are often reported as more accurate, particularly on small or densely packed objects, while one-stage detectors are substantially faster and better suited to real-time or resource-constrained deployment. H1 (Section 1.2) is a direct operationalisation of this framing for the road-damage task, predicting that YOLOv8 would trade a small amount of accuracy for a large gain in computational efficiency. Whether the accuracy side of that trade-off holds at the dataset scale used in this thesis is tested directly in Section 4.1.3, and is discussed in relation to this framing in Section 5.1.

2.3 Risk scoring and severity estimation for infrastructure

Pavement condition assessment has a long-established manual tradition, typified by indices such as the Pavement Condition Index, which combine observed distress type, severity, and extent into a single numerical condition score used for maintenance planning (Shahin, 2005). The risk-scoring model used in this thesis (Section 3.3) follows the same underlying logic, combining a fixed per-class severity weight with detection confidence into a single interpretable score, but is deliberately rule-based rather than learned, so that every score can be traced back to an explainable cause. This is a deliberate simplicity choice: a rule-based score is auditable by a non-technical stakeholder, such as a municipal maintenance planner, in a way that a learned scoring function is not.

2.4 Geospatial routing with cost-augmented graphs

Routing algorithms that penalise undesirable segments by adding a cost term to an otherwise standard shortest-path formulation are a long-standing approach in navigation and logistics research, used, for example, to route vehicles away from congestion, tolls, or known hazards rather than purely minimising distance or travel time. A directly analogous application outside the road-damage domain is Purchase and Breetzke (2026), who augment an OpenStreetMap-derived street network with a crime-risk cost term to compute safer, rather than strictly shortest, routes through a South African city, finding that the risk-weighted route was on average substantially longer than the shortest path but exposed travellers to roughly half the risk, a trade-off pattern this thesis's own H3 result (Section 4.1.5) closely mirrors for road-damage risk rather than crime risk. This thesis applies the same idea to road damage risk: OpenStreetMap road network data, accessed through OSMnx (Boeing, 2017), is augmented with a risk-based cost term per edge (Section 3.4), and a standard shortest-path algorithm is then run on the augmented graph exactly as it would be on the unmodified one. The novelty in this thesis is not the routing algorithm itself, which is deliberately kept simple and explainable, but the fact that the cost term is derived from real, automatically detected road damage rather than from static, pre-existing metadata such as road class or speed limit.

2.5 The gap this thesis fills

The great majority of road-damage detection literature stops at detection metrics: precision, recall, and mAP on a held-out test split of the same benchmark used for training. Comparatively lit-

tle published work connects detection output to a downstream decision, such as a maintenance priority ranking or a routing decision, and fewer still evaluate that downstream decision against independently collected, real-world ground truth rather than against the same benchmark's own labels. This thesis addresses that gap directly: every stage of the pipeline, from detection through risk scoring to routing, is evaluated against explicit, pre-registered hypotheses (H1–H4), and three of the four hypotheses are tested against ground truth that was collected independently of, and after, model training, rather than against a held-out split of the same training distribution.

3 Methodology

3.1 Dataset construction

All models are trained on subsets of RDD2022 (Arya et al., 2024), which provides 26 869 training, 5 758 validation, and 5 758 test images across seven national/collection sources (Japan, Norway, India, the United States, the Czech Republic, and two Chinese collection methods, drone- and motorbike-mounted). Only the four damage classes used throughout this thesis, longitudinal cracking (D00), transverse cracking (D10), alligator cracking (D20), and potholes (D40), are retained; a fifth, unofficial class present in the raw converted labels is filtered out at processing time.

Three processed datasets were built at increasing scale, summarised in Table 1: an initial small dataset used to validate that the full pipeline, from raw data to trained model to evaluation, worked end to end; a medium dataset used for a first scaled-up retrain; and a final, large dataset used for all results reported in Section 4. Progressing through these scales was necessary because auditing the small and medium datasets revealed two compounding biases inherited from RDD2022's raw class and country distribution. First, class D00 outnumbered classes D20 and D40 by a factor of two to three in both smaller datasets, despite D20 and D40 being the two highest-severity classes in the risk-scoring model (Section 3.3). Second, Japan accounted for approximately 46% of both smaller processed datasets, against approximately 28% of the raw pool, while the Czech Republic was represented by zero images in the small dataset's validation split.

The large dataset was constructed with an explicit balancing procedure rather than by simple random sampling. For each of the four classes, a fixed per-class image quota (1 100 training images, 150 validation, 150 test) was distributed as evenly as possible across every national source that contains at least one example of that class, using a water-filling allocation: countries are given an equal share of the quota, and any shortfall from a country with too few available images is redistributed across the remaining countries. This was necessary rather than optional, because auditing raw class-by-country availability found that class D40 has zero images from Norway, the United States, or the Czech Republic in RDD2022 at all; a fixed seven-way split would have been infeasible for this class, so the allocation is computed per class over only the countries that actually contain it. A fixed number of damage-free ("negative") images (200 training, 30 validation) was additionally included per split, so that the trained models also see undamaged road surface as background.

Table 1: Dataset-size iterations used in this thesis. All splits are drawn from RDD2022's own train/val/test partitions, filtered to the four damage classes used in this thesis.

Dataset	Train	Val	Test
Small	277	60	60
Medium	1 034	224	221
Large	4 291	597	592

3.2 Model architectures and training configuration

The two architectures were chosen to represent the two detector families compared in H1. YOLOv8n is the smallest variant of the current YOLO family and is designed for fast inference on modest hardware, which matches the deployment setting discussed in Section 5.6. Faster R-CNN is the canonical two-stage detector and a long-established reference point in object detection, providing a well-understood baseline for the two-stage side of the comparison. Both have COCO-pretrained weights, so the same starting point and training protocol can be applied to each.

Two object detection architectures are trained and compared, directly operationalising H1: YOLOv8n (Jocher et al., 2023), a lightweight, single-stage detector (approximately 3.0 million parameters), and Faster R-CNN with a ResNet50 backbone and a feature pyramid network (Ren et al., 2015), a two-stage detector (approximately 43.3 million parameters), initialised from COCO-pretrained weights in both cases.

Both architectures are trained identically at every dataset scale, so that any measured difference reflects the architecture rather than an incidental difference in training conditions: the same image resolution (640 px), the same train/validation/test split, and the same effective batch size are used for both models at a given dataset scale (Table 2). Because Faster R-CNN's memory footprint is substantially larger than YOLOv8n's at a given batch size, its effective batch size is reached through gra-

Table 2: Training configuration at the final (large) dataset scale, applied identically to both architectures unless noted.

Parameter	Value
Image size	640 px
Effective batch size	16
Maximum epochs	100
Early stopping patience	20 epochs (val. loss)
Optimiser	SGD
Mixed precision	Enabled (Faster R-CNN)
Gradient accumulation	4 steps (Faster R-CNN)

dient accumulation (a small number of images per forward/backward pass, with gradients summed over several steps before an optimiser update) rather than a single large batch, together with automatic mixed-precision training; both measures reduce peak GPU memory without changing the effective batch size, which was necessary given training was carried out on a single local GPU with 8 GB of memory. Both architectures use early stopping on validation loss (YOLOv8n) or a fixed patience on validation loss (Faster R-CNN), rather than a fixed epoch budget, so that neither model is arbitrarily under- or over-trained relative to the other.

3.3 Risk-scoring model

Each accepted detection (confidence at or above the detector's confidence threshold) is converted into a per-detection risk score as the product of a fixed, class-dependent base severity and a mild confidence multiplier:

$$r_i = s_{c_i} \cdot m(p_i), \tag{1}$$

where r_i is the risk score of detection i, c_i is its damage class, s_{c_i} is that class's fixed base severity (Table 3), and p_i is its confidence. The confidence multiplier is defined as

$$m(p) = \begin{cases} m_{\min} & p \le \tau \\ m_{\min} + \frac{p-\tau}{1-\tau}(1 - m_{\min}) & \tau < p < 1 \\ 1 & p \ge 1, \end{cases} \tag{2}$$

with confidence floor $m_{\min} = 0.7$ and confidence threshold $\tau = 0.25$ (the same threshold at which a detection is accepted at all). The multiplier is deliberately narrow, spanning only $[0.7, 1.0]$, rather than the full $[0, 1]$ range: a detection's confidence reflects how sure the model is that the damage is present, not how physically dangerous it is once confirmed, so a low-confidence real pothole should not be scored as though it were only mildly hazardous.

Table 3: Fixed base severity per damage class.

Class	Base severity s_c
D00 – Longitudinal crack	1.0
D10 – Transverse crack	1.2
D20 – Alligator crack	2.0
D40 – Pothole	3.0

Bounding-box size is deliberately excluded from the severity calculation, which is a departure from the original proposal wording of H2 ("damage type *and size*"). In a single two-dimensional photograph, the apparent size of a bounding box is a function of camera distance and angle at least as much as it is a function of the real-world size of the damage; using it as a severity multiplier would let camera technique, rather than the underlying road condition, distort the risk score. This deviation is stated explicitly here, and its consequences for H2 are discussed in Section 5.3.

Multiple detections, whether within a single photograph or matched to the same road segment, are combined with the same rule at both levels:

$$R = \max_i(r_i) + \beta \sum_{j \neq \arg\max_i(r_i)} r_j, \tag{3}$$

with damping factor $\beta = 0.3$. The single worst detection dominates the combined score, so that one severe hazard is not diluted by several minor ones, while additional co-occurring damage still increases the score rather than being ignored. At image level, the combined score R is normalised to $[0, 1]$ by dividing by a ceiling equal to the highest base severity (3.0, so that one confidently detected pothole alone already saturates the score), and categorised as low (< 0.34), medium (0.34–0.67), or high (≥ 0.67).

A combined risk score should satisfy three criteria: (i) the single worst hazard should dominate the score; (ii) additional damage should still increase the score; and (iii) confidence should only nudge the score within a narrow range, rather than deciding it. To illustrate how alternative combination rules meet these criteria, consider the photograph Test_Route_16 (Figure 2), processed by the deployed pipeline. It contains five accepted detections: one transverse crack (D10, confidence 0.587, per-detection risk 1.002) and four longitudinal cracks (D00, with confidences 0.754, 0.427, 0.402 and 0.372, and per-detection risks 0.902, 0.771, 0.761 and 0.749). Four scoring rules applied to these detections give the following results on the 0–1 scale:

- **Severity only** (highest base severity present, 1.2): $1.2/3.0 = 0.40$, medium. The four longitudinal cracks and all confidence information are ignored.

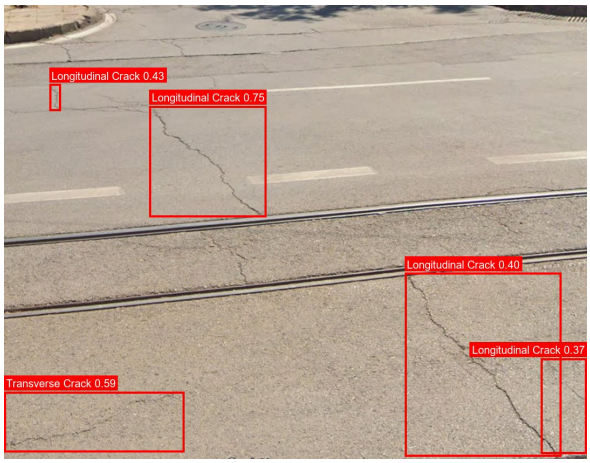


Figure 2: Test_Route_16 with the five accepted detections used in the worked example. Red boxes show the detector's output; each label gives the damage type and confidence.

- **Sum of base severities, no confidence**: 5.2/3.0 saturates at 1.0, high. Five modest cracks dominate the score.
- **Highest confidence only** (0.754): 0.75, high. The class of each detection, and the other four detections, are ignored.
- **Current rule** (Equations 1–3): $1.002 + 0.3 \times 3.183 = 1.957$, normalised to 0.652, medium.

Against these criteria, severity-only fails (ii) and (iii): the four longitudinal cracks have no effect on the score, and confidence has no effect at all. Summing severities fails (i), since five modest cracks outweigh the worst one, and fails (iii), since confidence is ignored. Highest confidence only fails (i), since the most severe damage does not decide the score, and fails (iii), since confidence decides the whole score rather than nudging it. The current rule meets all three: the transverse crack contributes 1.002 of the 1.957 total (51%), the four cracks together add 0.955, and confidence moves each longitudinal crack only within the multiplier range $[0.7, 1.0]$ (0.749 at confidence 0.372, 0.902 at 0.754). The example illustrates how each rule behaves against these criteria; it is not evidence that the current rule is more accurate than the alternatives, since H2 is not supported at its target threshold (Section 4.1.4). The resulting score of 0.652 also lies close to the medium/high boundary of 0.67, so a small change in the rule would change this category.

3.4 Geospatial matching and risk-aware routing

The routing and geospatial-matching components of the pipeline are scoped to a single real pilot area, Sofia, Bulgaria, rather than the whole of Bulgaria, on the grounds that the intended use case is a municipal tool and a city-scale road network is both sufficient to demonstrate the pipeline and small enough to download and route over efficiently. The Sofia drivable road network (an 8 km radius around the city centre, 11 363 nodes and 27 429 edges) is obtained from OpenStreetMap via OSMnx (Boeing, 2017) and cached locally.

Each detection's GPS coordinate is matched to the nearest edge in this graph, identified by its two endpoint node identifiers and a key disambiguating parallel edges between the same two nodes. A match is accepted only if the nearest edge is within a fixed distance threshold (50 m); points further away, or outside the graph's bounding box entirely, are left unmatched rather than force-matched to a plausible but potentially incorrect road. All detections matched to the same edge are combined into a segment-level risk using Equation 3 applied over that edge's detections, producing two figures: an unnormalised *total risk*, used for maintenance-priority ranking, and a *risk density*, the total risk divided by the segment's real-world length, used for routing, so that a long segment is not penalised more heavily purely for having had more opportunities to be photographed.

The risk-aware routing algorithm adds a risk-based cost term to each edge's physical length:

$$c(e) = \ell(e) + \alpha \cdot d(e), \tag{4}$$

where $c(e)$ is the edge's routing cost, $\ell(e)$ its length in metres, $d(e)$ its risk density (zero for edges with no matched detections), and α a weighting constant. A standard shortest-path algorithm is then run twice on the same graph: once using $\ell(e)$ alone (the "shortest" route), and once using $c(e)$ (the "risk-aware" route). The value $\alpha = 15\,000$ was set empirically against the available pilot data: below this value, the extra distance cost of detouring around the single highest-risk known segment outweighed its risk penalty and the shortest and risk-aware routes coincided; at this value, avoiding that segment became worthwhile. This is a rule-based prototype value rather than a calibrated safety threshold, and is exposed as an adjustable parameter in the deployed application rather than fixed in code.

3.5 Prototype implementation

The deployed system is a web prototype built with Streamlit (version 1.62), with interactive maps rendered using Folium. It consists of four pages that follow the order of the pipeline. Screenshots of each page are provided in Appendix D.

- **Dashboard** summarises all stored observations: the number of processed images, detected damages, affected road segments, the average risk score, and the number of high-

Table 4: Tables in the local SQLite database.

Table	Stored content (one row per)
`observations`	saved image: file hash (unique, so duplicate uploads are rejected), GPS coordinate and its source (EXIF, manual, or map selection), upload time, model version, confidence threshold, inference time, observation risk score and category, and matched road edge with matching distance
`detections`	accepted detection: damage class, confidence, bounding box, and risk contribution
`road_segments`	matched road edge: aggregated risk, risk density, risk category, and number of observations
`routes`	route comparison: origin, destination, α used, shortest and risk-aware distance and exposure, and distance increase

risk locations, together with detections per damage class, observations per risk category, and an overview map.

- **Upload and Detect** accepts a geotagged street photograph, runs the detector, and displays the annotated image, each accepted detection with its confidence and risk contribution, and the observation risk. An observation is stored only when the user confirms it. Each file is identified by a SHA-256 hash of its contents, so re-uploading an identical image is recognised.

- **Damage Map** shows every stored observation as a marker and the road segments with recorded damage, coloured by segment risk, with an optional risk heatmap layer.

- **Safer Route Planner** takes an origin and a destination on the road network, computes the shortest and the risk-aware route, and reports their distance and hazard exposure. The risk-aversion weight α (Equation 4) is adjustable on this page.

Detection, risk scoring, routing, and storage run locally. All records are held in a single SQLite database file, and uploaded images are kept in a local upload folder. The database has four tables, summarised in Table 4, and the road network is loaded from the cached OpenStreetMap graph described in Section 3.4. The only external data loaded by the application is the map tile imagery.

Recording the model version, confidence threshold, and matched road edge with each observation allows any displayed result to be traced back to the configuration that produced it. The 10% distance limit used in H3 is applied in the offline analysis: the application records the distance increase of each route but does not enforce the limit.

Figure 3 shows the workflow from the user's perspective, with one lane for each group of actors.

3.6 Manual data collection and annotation

Two separate sets of manually collected road photographs were used in this thesis, for two different purposes.

The first set comprises 201 road-damage photographs, independently collected and manually annotated by the author for the four damage classes used throughout this thesis. Images were obtained by mounting a smartphone camera inside the car and recording continuous video while driving on public roads around Sofia, outside the Sveta Troitsa area subsequently used for the routing and heatmap-validation evaluations (Sections 4.1.5 and 4.1.6), so that the images used for fine-tuning and the images used for those two evaluations do not overlap. Individual frames containing visible damage were then extracted from the recorded video and annotated using CVAT (Computer Vision Annotation Tool), exported in the Ultralytics YOLO Detection 1.0 format (normalised bounding boxes, class identifiers 0–3 matching the same convention used for RDD2022). No single fixed camera angle was imposed: because frames were sampled from continuous driving footage, the resulting images naturally cover a range of distances and viewing angles to the road surface, which was intentional, so that the fine-tuned models would not be trained on an artificially narrow viewpoint and would generalise better to images captured from different angles at deployment time. Every image/label pair in this set was verified programmatically before use: that every image has a corresponding label file and vice versa, that every label file contains only valid class identifiers and in-range normalised coordinates, and that the damage class indicated in each file's name is actually present among its own annotated classes; all 201 pairs passed these checks. The resulting class distribution is imbalanced (150, 75, 90, and 35 images containing at least one D00, D10, D20, and D40 instance respectively), which is discussed as a limitation in Section 5.5. None of the 201 images retain GPS metadata, since frame extraction from video does not preserve the per-frame location information a native photo capture would. Because of this, the set was used for the domain-adaptation fine-tuning and held-out evaluation described below, which do not require location information, but not for the geospatial matching and routing evaluation (Sections 3.4, 4.1.5 and 4.1.6), which do.

Examples of annotated frames from this set are shown in Figure 4.

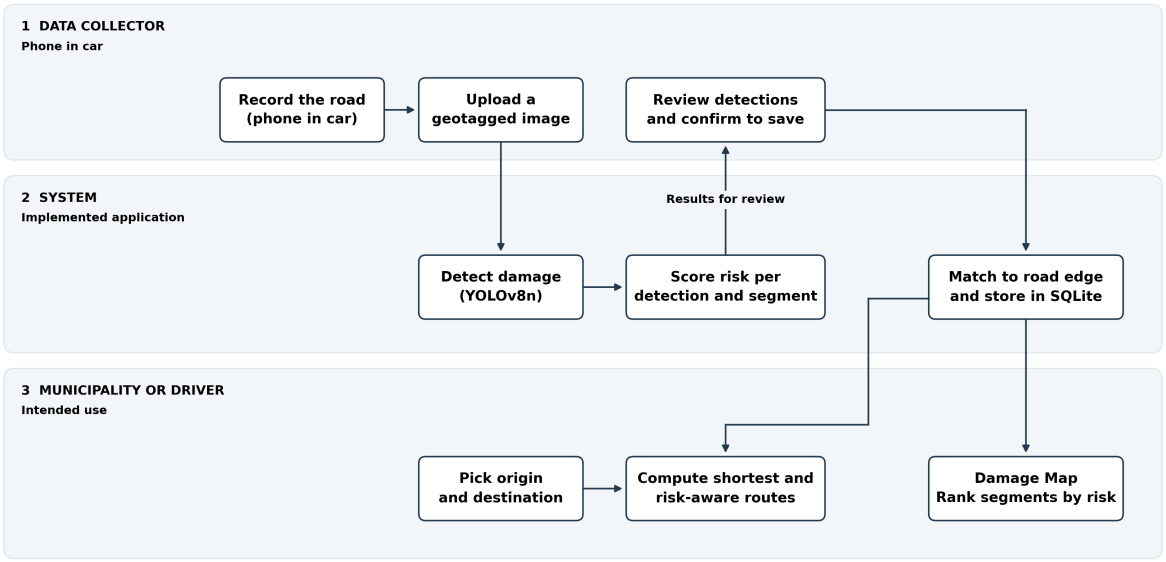


Figure 3: User-perspective workflow. A data collector uploads a geotagged photograph; the system detects damage, scores it, and stores the observation after the user confirms it. Municipal staff use the Damage Map to rank road segments, and drivers pick an origin and destination to compare routes. The first two lanes describe the implemented application, with recording done on a phone outside it; the bottom lane describes intended use.

The second set comprises 17 photographs collected specifically to support the routing and heatmap-validation evaluations (Sections 4.1.5 and 4.1.6), for which real coordinates were required. These were taken with the same smartphone as the first set, but not while driving: the device's standard camera application was used to photograph individual sections of damage that were judged worth recording within the neighbourhood, rather than extracting frames from continuous video. Location services were enabled so that GPS coordinates were retained in each image's EXIF metadata. Unlike the first set, this set was deliberately restricted to within a single pilot neighbourhood, Sveta Troitsa, Sofia, covering its main streets; the neighbourhood's evaluation boundary was subsequently derived directly from the actual GPS extent of these photographs, rather than being predefined. Unlike the first set, these images were not manually annotated; they were instead processed directly through the same detection and risk-scoring pipeline used by the deployed application, so that the risk data they contribute reflects the system's own detections rather than human-labelled ground truth.

3.7 Domain-adaptation fine-tuning

A model trained exclusively on RDD2022 is not guaranteed to generalise to a specific deployment area: RDD2022's images were collected under capture protocols, camera hardware, and road/lighting conditions specific to its seven source countries, none of which is Bulgaria. To test and address this, 201 photographs were independently collected and manually annotated in the pilot area (Sveta Troitsa, Sofia), verified for label consistency, and split into three non-overlapping subsets (Table 5): a fine-tuning training set, a small fine-tuning validation set used to monitor for overfitting during fine-tuning, and a held-out test set that neither model was trained or fine-tuned on at any stage, used exclusively for the before/after comparison in Section 4.1.2.

Table 5: Manual Sofia dataset split used for domain adaptation.

Split	Images
Fine-tuning train	131
Fine-tuning validation	28
Held-out test	42

Both large-dataset-trained models were fine-tuned on the 131-image fine-tuning split, starting from their RDD2022-large weights, using a reduced learning rate relative to training from scratch and a short epoch budget with aggressive early stopping. The intent is a targeted adjustment of existing weights toward the pilot area's visual conditions, not a retrain: 131 images is a small fraction of the 4 291-image RDD2022-large training set, and a reduced learning rate limits how far the already-learned representations can move.

3.8 Evaluation and statistical analysis plan

Detection performance is reported using precision, recall, F1, mean average precision at an IoU threshold of 0.5 (mAP50), and averaged over IoU thresholds from 0.5 to 0.95 (mAP50-95), computed identi-

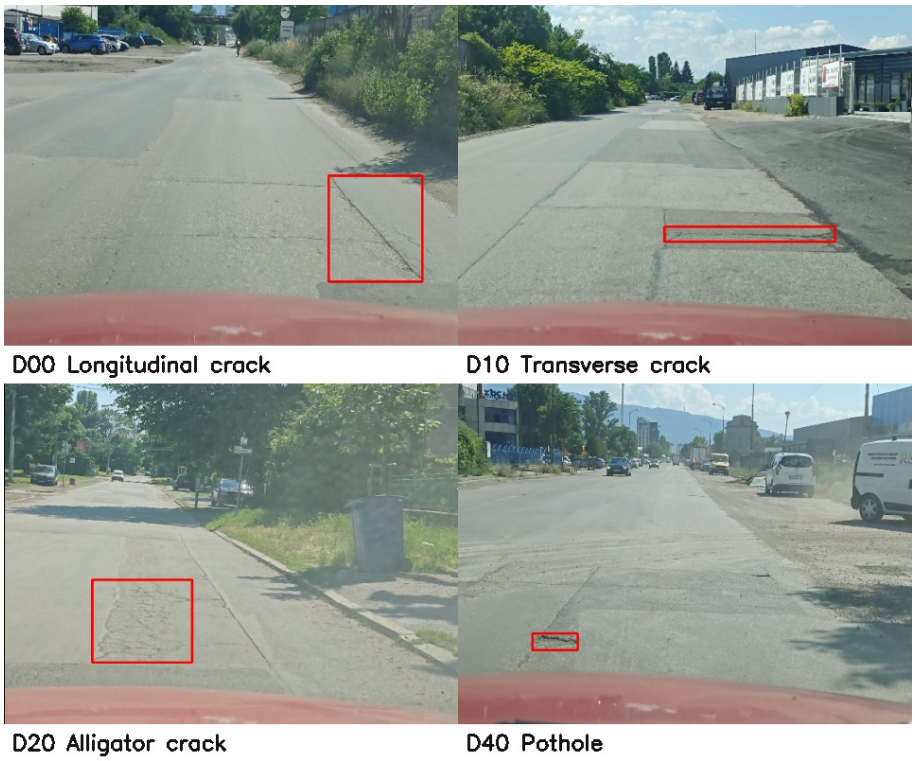


Figure 4: Example annotations from the manually collected Sofia dataset for the four damage classes used in this thesis, with ground-truth bounding boxes in red. Images are shown for illustration only; they are drawn from the same set used for fine-tuning and held-out evaluation.

cally for both architectures on the same test splits.

Each metric answers a different question. Precision is the fraction of predicted damage that is correct, and recall is the fraction of real damage that is detected. F1 combines the two into a single value, so that models can be ranked on one number. Object detection also requires that a predicted box overlaps the ground-truth box. Intersection over Union (IoU) measures this overlap as the area of the intersection divided by the area of the union, and a detection counts as correct when its IoU reaches a threshold. mAP50 uses a threshold of 0.5, a lenient localisation criterion, while mAP50-95 averages over thresholds from 0.5 to 0.95 and so rewards accurate box placement. mAP is the standard summary metric for object detection. It is reported alongside F1 because the RDD2022 challenge reports F1 (Arya et al., 2022a), which allows the results to be compared with that benchmark.

H1's two criteria are each tested with a statistical procedure suited to the specific claim being made, rather than a single generic test. The complexity criterion (lower training and inference cost) is tested with a paired Wilcoxon signed-rank test on per-image inference time, measured for both architectures on the same set of test images in the same session; the Wilcoxon test is used, rather than a paired t-test, because inference-time distributions are typically right-skewed rather than normally distributed. The accuracy criterion (mAP50-95 within two percentage points) is an equivalence claim, not a claim of zero difference, so it is tested with a bootstrap confidence interval on the mAP50-95 difference between the two architectures (300 resamples of the test set, with replacement), following the logic of two one-sided equivalence tests: H1's accuracy criterion is considered supported only if this interval falls entirely within ± 2 percentage points, rather than by inspecting whether a single point-estimate difference happens to be small.

H2 is tested by comparing the deployed pipeline's automatically generated low/medium/high risk category against an independently assigned ground-truth label for the same images, using accuracy, per-category precision/recall/F1, and Cohen's κ; κ is reported alongside raw accuracy because, with three categories, a nontrivial level of agreement is expected by chance alone, and κ corrects for this. The pre-registered target for H2 is a macro-averaged F1 of at least 0.75.

H3 is tested by generating a fixed number of independent origin-destination pairs, comparing the shortest and risk-aware route for each, and applying a paired Wilcoxon signed-rank test to the hazard exposure of the two routes across the sample. Hazard exposure of a route is the sum, over every road segment the route traverses, of that segment's total risk (Equation 3, applied to

all detections matched to the segment); segments without recorded risk contribute zero. Exposure reduction is the percentage by which the risk-aware route's exposure falls below the shortest route's. The measure reflects both how many damaged segments a route passes and how severe they are, so avoiding one high-risk segment counts for more than passing several minor ones. Because the routing cost and the exposure measure are both computed from the same recorded risk, H3 shows that the risk-aware route avoids recorded damage, not that it avoids all damage in the area. The same non-parametric justification applies as for the complexity criterion above, and is additionally appropriate given that many routes are expected to have identical exposure on both sides (no hazard on the shortest path at all). The hypothesis asks for a "reasonable" increase in route distance; this is operationalised as an increase of at most 10% over the shortest route, and a route satisfies the constraint if it meets this limit.

H4 is tested with Spearman's rank correlation coefficient between independently assigned segment risk ratings and the pipeline's generated segment-level risk. Spearman's ρ, rather than Pearson's r, is used because the manual ratings are an ordinal scale with a small number of distinct values, for which a rank-based correlation is the more appropriate measure of monotonic association. H4 is supported if ρ is at least 0.5 and p is below 0.05.

In every case, ground truth used for H2 and H4 was assigned by looking at the image or road segment in question alone, before the model's output was consulted, so that the comparison is not biased by anchoring the manual judgement to the model's answer.

3.9 Qualitative usability evaluation

The hypothesis tests above are quantitative and evaluate the pipeline's outputs against ground truth; they do not speak to how a human user experiences or trusts the system, which is a separate and complementary question. To address this, a small qualitative usability evaluation was conducted, following the Research Quality Guidelines' allowance for qualitative methods, such as structured feedback and content analysis, to complement quantitative results.

Ten participants each completed the same short, guided walkthrough of the deployed application: uploading a photograph and reviewing its detection result, exploring the Damage Map, comparing a shortest route against a risk-aware route in the Safer Route Planner, and explaining, unprompted, what the displayed risk score and colour coding meant to them. Each participant then completed a short survey consisting of five closed, five-point Likert-scale questions covering ease of understanding, intuitiveness of the risk categories, map clarity, trust in the system's risk assessment, and perceived usefulness of the route comparison, followed by four open-ended questions on confusion points, trust, desired improvements, and general comments. The closed questions are analysed with descriptive statistics (mean, median); the open-ended responses are analysed qualitatively by grouping them into recurring themes, following standard content-analysis practice, with representative quotations retained for illustration. This design was chosen specifically because the trust and clarity questions map directly onto H2 and H4: H2 tests whether the risk-scoring rule is quantitatively accurate, while this survey tests whether users independently perceive that same risk information as trustworthy and clear, a distinct but related question that the quantitative hypothesis tests cannot answer on their own.

Table 6: Summary of hypothesis tests.

Hypothesis	Criterion	Result	Verdict
H1	mAP50-95 gap within ± 2 pp; lower inference time	CI outside margin; time $p=6.04\times10^{-99}$	Partially supported
H2	Macro F1 $\geq$ 0.75	Macro F1 0.582	Not supported
H3	Lower exposure; distance increase $\leq 10\%$	p=0.0017; 38/40 within limit	Supported
H4	$\rho \geq 0.5$, $p < 0.05$	ρ=0.683 (p=0.0036)	Supported

4 Results

4.1 Quantitative results

Quantitative results were collected from four distinct evaluation procedures, each matched to the hypothesis it tests (Section 3.8): automated detection metrics (precision, recall, F1, mAP) computed on each dataset scale's own RDD2022 test split and, separately, on the held-out Sofia test set before and after fine-tuning; a paired statistical comparison of the two architectures on shared test data; a manually labelled sample of 42 Sofia images for risk-classification agreement (H2); a systematically generated sample of routing comparisons (H3); and a manually rated sample of road segments correlated against generated segment risk (H4). Each is reported in turn below. Table 6 summarises the four hypothesis tests, and the criteria are defined in Section 3.8.

4.1.1 Detection performance across dataset iterations

Table 7 reports detection performance for both architectures at all three dataset scales, on each

scale's own RDD2022 test split. Both architectures improve monotonically and substantially as the dataset grows: YOLOv8n's mAP50-95 increases from 0.119 at the small scale to 0.300 at the large scale, and Faster R-CNN's from 0.043 to 0.186. YOLOv8n outperforms Faster R-CNN at every scale on every reported metric except precision at the medium scale, where Faster R-CNN is higher (0.543 versus 0.503).

At the large scale, per-class mAP50-95 (Table 8) shows that D40 (pothole), the class targeted most directly by the class-balancing procedure (Section 3.1) and the highest-severity class in the risk-scoring model, is the strongest class for both architectures, a reversal of the pattern observed before balancing was introduced.

4.1.2 *Domain-adaptation results*

Evaluated on the RDD2022 large-dataset test split, YOLOv8n and Faster R-CNN reach mAP50-95 of 0.300 and 0.186 respectively (Table 7). Evaluated instead on the held-out Sofia test set, without any fine-tuning, both collapse to approximately the same low level: 0.041 for YOLOv8n and 0.040 for Faster R-CNN, a relative drop of approximately 86% and 78% respectively (Table 9). Both architectures collapsing to approximately the same absolute performance despite very different RDD2022-native performance is consistent with a genuine domain-shift effect rather than a weakness specific to one architecture.

Fine-tuning both models on the 131-image fine-tuning split and re-evaluating on the same held-out Sofia test set produces a substantial improvement for both architectures (Table 10): YOLOv8n's F1 improves from 0.189 to 0.347 (a relative increase of approximately 83%), and Faster R-CNN's from 0.218 to 0.311 (approximately 43%). YOLOv8n's absolute mAP50-95 gain (+0.107) is approximately five times Faster R-CNN's (+0.021). Neither model's fine-tuned Sofia performance reaches its RDD2022-native performance. The fine-tuned YOLOv8n achieves a recall of 0.335 on this held-out test set.

4.1.3 *H1 – model comparison*

H1's two criteria were tested independently on the large-dataset models (Table 11). The accuracy criterion, that YOLOv8n's mAP50-95 would fall within two percentage points of Faster R-CNN's, is rejected: a 300-resample bootstrap over the shared test set gives a 95% confidence interval of $[+0.086, +0.117]$ on the YOLOv8n$-$Faster R-CNN mAP50-95 difference, entirely outside the ± 0.02 equivalence margin. The complexity criterion is supported: a paired Wilcoxon signed-rank test on per-image inference time (median 22.25 ms for YOLOv8n versus 307.33 ms for Faster R-CNN, measured on the same 592 test images in the same session) gives $p = 6.04 \times 10^{-99}$. Training wall-clock time shows the same pattern: 1.36 hours for YOLOv8n (100 of 100 epochs, no early stop) against 4.16 hours for Faster R-CNN (22 of 100 epochs, early stopped), on identical hardware.

Across every dataset scale, Faster R-CNN's lowest validation loss was reached within the first two to four epochs, with validation loss then increasing for the remainder of training; at the large scale, the best checkpoint was reached at epoch 2 of 22 completed epochs. This pattern persisted even at the largest dataset scale used.

4.1.4 *H2 – risk classification*

The deployed pipeline's automatically generated risk category was compared against an independently assigned ground-truth label for the 42-image held-out Sofia test set (Table 12). Macro-averaged F1 is 0.582, below H2's target of 0.75; H2 is not supported. Accuracy is 0.595 and Cohen's κ is 0.377, indicating fair-to-moderate agreement beyond chance.

Per-category results (Table 13) show precision of 0.91 on the high-risk category, against recall of only 0.53: when the pipeline assigns a high-risk category, it is almost always correct, but it misses a substantial share of images that were independently judged high-risk. Of 19 images rated high-risk, nine were not scored high: eight were scored down to medium and one to low. The nine missed images comprise two pothole-labelled, three alligator-crack-labelled, one transverse-crack-labelled and three longitudinal-crack-labelled images. In an earlier inspection of the detector's raw output for three of these cases, the relevant damage was not detected at all rather than detected and undervalued: two pothole-labelled images returned only lower-severity classes (alligator and longitudinal cracking) or none of the pothole instances present, and one alligator-crack-labelled image returned zero detections.

4.1.5 *H3 – routing evaluation*

An initial sample of 40 independent origin-destination pairs, drawn at random from the full Sofia road graph (27 429 edges), was compared under shortest-path and risk-aware routing. At the time of this sample, only 4 edges in the entire graph carried any recorded risk, and zero of the 40 generated routes crossed one of them; no paired test could be computed from this sample, since shortest-path and risk-aware exposure were identical (both zero) for every route.

A second sample was then generated after increasing real detection coverage within one neighbourhood, Sveta Troitsa: 14 additional geotagged photographs were collected and processed through the deployed pipeline, joining three already-processed photographs from the same area, bring-

Table 7: Detection performance across dataset-size iterations, evaluated on each scale's own RDD2022 test split.

Model	Dataset	Precision	Recall	F1	mAP50	mAP50-95
YOLOv8n	Small	0.368	0.305	0.334	0.297	0.119
YOLOv8n	Medium	0.503	0.434	0.466	0.424	0.194
YOLOv8n	Large	**0.643**	**0.496**	**0.560**	**0.548**	**0.300**
Faster R-CNN	Small	0.227	0.205	0.216	0.136	0.043
Faster R-CNN	Medium	0.543	0.300	0.387	0.340	0.148
Faster R-CNN	Large	0.434	0.379	0.404	0.373	0.186

Table 8: Per-class mAP50-95 on the large-dataset RDD2022 test split.

Class	YOLOv8n	Faster R-CNN
D00 – Longitudinal crack	0.211	0.131
D10 – Transverse crack	0.226	0.164
D20 – Alligator crack	0.316	0.169
D40 – Pothole	**0.448**	**0.277**

Table 9: mAP50-95 on the RDD2022 test set versus the held-out Sofia test set, before any fine-tuning.

Model	RDD2022	Sofia (no FT)
YOLOv8n	0.300	0.041
Faster R-CNN	0.186	0.040

Table 10: Held-out Sofia test set performance, before and after domain-adaptation fine-tuning.

Model	Stage	F1	mAP50-95
YOLOv8n	No fine-tuning	0.189	0.041
YOLOv8n	Fine-tuned	**0.347**	**0.148**
Faster R-CNN	No fine-tuning	0.218	0.040
Faster R-CNN	Fine-tuned	0.311	0.061

Table 11: H1 verdict, large-dataset models.

Criterion	Result	Verdict
Accuracy (mAP50-95 within 2 pp)	95% CI $[+0.086, +0.117]$	Not supported
Complexity (train + inference time)	p=6.04×10^{-99}	Supported

ing the number of distinct road segments with recorded risk from 4 to 16, all within or adjacent to this neighbourhood. Restricting route sampling to the 206 graph nodes located inside this neighbourhood's bounding box, 40 fresh independent routes were generated and compared (Table 14). Of these, 18 (45%) crossed a segment with recorded risk; among those, mean and median hazard exposure reduction were 45.6% and 37.2% respectively, and 95% of all 40 routes satisfied a 10% maximum distance-increase constraint. A paired Wilcoxon signed-rank test on hazard exposure across all 40 routes gives $p = 0.0017$; H3 is supported.

4.1.6 H4 – heatmap validation

The same 16 Sveta Troitsa road segments used in Section 4.1.5 were independently rated on a 1–5 scale and compared against the pipeline's generated segment-level risk (Table 15). Manual ratings correlate significantly with both risk measures: Spearman's $\rho = 0.683$ ($p = 0.0036$) against total (aggregated) risk, and $\rho = 0.558$ ($p = 0.0247$) against risk density. Both exceed H4's target of $\rho \geq 0.5$ with $p < 0.05$; H4 is supported.

4.2 Qualitative results

Qualitative results were collected via the guided walkthrough and survey described in Section 3.9: five closed, five-point Likert questions and four open-ended questions, completed after a short hands-on task walkthrough of the deployed application. Ten participants ($n = 10$) completed the walkthrough and survey, none with prior experience of a similar road-condition or navigation application. Four of the five closed-question means were 4.2 or higher on the five-point scale (Table 16). The risk-category and map-clarity questions scored highest (both 4.6), and the trust question lowest (3.8, minimum 3), though still on the positive side of the scale.

Thematic analysis of the open-ended responses identified seven recurring themes.

Accurate mental model of the risk colour-coding. All ten participants described the green, orange or yellow, and red severity mapping correctly in outline, before any explanation was given in the survey. Three participants (P03, P04, P10) described the score as combining several detections rather than a single value, closely matching the actual aggregation rule (Equation 3): "I think based on all the damage on the road segment the risk is combined and multiplied" (P03). P10's wording ("summed risk") approximates this rule, which in fact takes the largest detection and adds a damped share of the rest.

Interface and control clarity issues. Seven of ten participants reported something unclear in the interface, from button visibility and marker selection to the meaning of the controls. Three of them (P03, P04, P09) named the Safer Route Planner specifically, for example its risk-aversion (α) slider

Table 12: H2 overall results.

Metric	Value
Accuracy	0.595
Cohen's κ	0.377
Macro F1 (target $\geq$ 0.75)	0.582

Table 13: H2 per-category results.

Category	Precision	Recall	F1
Low	0.50	0.50	0.50
Medium	0.48	0.73	0.58
High	**0.91**	0.53	0.67

and pin-selection map: "it took me some time to find where the map for choosing pins directly on it is" (P04). The underlying routing mechanism was independently supported quantitatively (Section 4.1.5); this result indicates a separate, addressable interface clarity issue rather than a problem with the mechanism itself.

Onboarding and explanation are needed. Four participants (P03, P05, P07, P09) asked for guidance on what the controls and codes mean, including the alpha setting, the damage class codes, and the thresholds: "The application will be more useful if it has something like a guide with deep explanations what is the function for every thresholds that can be set" (P03).

Trust is conditional, not unconditional. Every participant said they would use or consider the system, but six of ten attached a condition. Two conditioned trust on detection accuracy: one linked it to the per-class detection confidence, "it depends on the thresholds set... will be useful for the classes that have highest confidence" (P03), and another noted that incorrect detections would affect confidence in the recommended route (P08). Others conditioned trust on how recent the imagery is (P06, P10), on being able to inspect the supporting images (P07), or on the risk-aware route adding only a small amount of distance (P09). The accuracy conditions corroborate, from a non-technical perspective, the detection-recall limitation identified quantitatively in Section 5.3.

Data currency and repair status. Four participants (P01, P06, P08, P10) raised how up to date the information is: whether a segment has been repaired, when the images were collected, and whether an unmarked road had been checked. P01 and P08 asked for a way to mark repaired segments, and P08 added that such reports should be checked before the map is updated.

Participants envision real-time, mobile deployment. Four participants independently proposed moving from the current batch-upload prototype toward real-time, in-vehicle use: GPS-linked proximity notifications (P02, P05, P10), a

Table 14: H3 results, Sveta Troitsa sample (40 routes).

Metric	Value
Routes crossing a recorded-risk segment	18/40 (45%)
Mean / median distance increase (all routes)	1.50% / 0.00%
Routes within the 10% distance constraint	38/40 (95%)
Mean / median exposure reduction	45.6% / 37.2%
Wilcoxon signed-rank test	p=0.0017

Table 15: H4 results, 16 Sveta Troitsa segments.

Comparison	Spearman ρ	p
vs. total risk	**0.683**	0.0036
vs. risk density	0.558	0.0247

mobile interface like typical routing maps (P05), and automatic ingestion from a continuously recording dashcam (P01).

The prototype was correctly read as a prototype. Two participants (P03, P04) explicitly framed their feedback in prototype terms: "for a prototype app it's also really good... showing that everything is working perfectly" (P03).

Full response data, descriptive statistics, and the complete thematic coding with all supporting quotations are provided in Appendix C.

5 Discussion

5.1 Interpreting H1

YOLOv8n outperforms Faster R-CNN on every measured axis at the dataset scale used in this thesis: higher accuracy at every dataset size, faster training, and faster inference. H1 predicted a small accuracy trade-off in exchange for lower computational cost, consistent with the conventional one-stage/two-stage framing reviewed in Section 2.2; the complexity half of that framing is strongly confirmed, but the accuracy half is not merely unconfirmed, it is reversed; the two-stage detector is both slower and less accurate here, rather than more accurate and slower. A plausible explanation lies in model capacity relative to available data: Faster R-CNN's approximately 43 million parameters are roughly fourteen times YOLOv8n's approximately 3 million, and its validation loss began rising within two to four epochs at every dataset scale tested, including the largest (Section 4.1.3). This is the signature of overfitting a large-capacity model to a training set that, despite being the largest built in this thesis, may still be small relative to what a 43-million-parameter two-stage detector needs to reach its full accuracy potential. This does not indicate that Faster R-CNN is inherently less suit-

Table 16: Qualitative usability survey, closed questions ($n = 10$).

Question	Mean	Median
Detection results easy to understand	4.2	4.0
Risk score/category intuitive	**4.6**	5.0
Map made risky roads clear	**4.6**	5.0
Trust the risk assessment	3.8	4.0
Route comparison useful	4.4	4.0

able for this task, only that, under the data volume and training budget available here, it is the weaker practical choice, a conclusion specific to this dataset scale rather than a general claim about the architecture.

5.2 The domain gap and what it implies

The single largest effect observed in this thesis is not the difference between the two architectures, but the difference between each architecture's performance on RDD2022 and its performance on real, previously unseen photographs from the pilot deployment area. Both YOLOv8n and Faster R-CNN, despite reaching very different absolute performance on RDD2022 (Section 4.1.3), collapse to approximately the same low performance on the Sofia test set before any adaptation (Section 4.1.2). A model that scores well on an established benchmark is not, by that fact alone, evidence that it will work in a specific deployment location; this motivated the deployment-locality argument set out in Sections 1.1 and 3.7, and this result confirms that the concern was not merely theoretical.

The more encouraging finding is how much of that gap a small amount of local data closes. Fine-tuning on 131 photographs, a little over 3% of the large RDD2022 training set, produced a relative F1 improvement of 83% for YOLOv8n and 43% for Faster R-CNN on the same held-out Sofia images. This suggests that domain adaptation for this kind of task does not require a large local data-collection effort to produce a meaningful, measurable improvement, which is a practically important finding for any organisation considering deploying a similar system in a new city: a modest, low-cost local data-collection and fine-tuning step is worth doing even before considering whether to collect a large local training set from scratch.

5.3 H2's shortfall, explained

H2 is not supported at its target threshold, but the per-category results in Section 4.1.4 point to a specific, identifiable cause rather than a general weakness of the risk-scoring rule. Precision on the high-risk category is 0.91: when the pipeline does assign a high-risk label, it is almost always right. The shortfall is concentrated entirely in recall. In the three cases inspected, the cause was a detector miss rather than a correctly detected pothole or alligator crack being scored as minor; the remaining six missed images were not inspected at the level of individual detector output, so this is not claimed for them. Because the risk-scoring rule (Section 3.3) can only score what the detector actually reports, a recall gap in detection necessarily propagates into a recall gap in risk classification. This reframes H2's result: it is evidence of the same detection-recall limitation quantified for Sofia in Section 4.1.2 (fine-tuned recall of 0.335), observed again downstream, rather than an independent flaw in the combination rule in Equation 3. It follows that improving detection recall on real Sofia photographs, rather than redesigning the scoring rule, is the more direct route to improving H2's result. Independent, qualitative support for this same conclusion emerged from the usability study (Section 4.2): one participant spontaneously identified detection confidence, rather than the scoring logic, as the limiting factor on their own trust in the system, arriving at the same diagnosis as the quantitative analysis from an entirely independent, non-technical angle.

5.4 H3 and H4 – the coverage argument

The initial, city-wide attempt at testing H3 produced no usable result, not because the risk-aware routing mechanism failed, but because only four of the graph's 27 429 edges carried any recorded risk at that point, so a randomly sampled route almost never had anything to avoid. Concentrating further data collection into a single neighbourhood, rather than spreading it thinly across the city, increased the number of risk-carrying segments from four to sixteen and made both H3 and H4 statistically testable, with both hypotheses subsequently supported (Sections 4.1.5 and 4.1.6). Read together, the two attempts at H3 demonstrate that the routing and risk-aggregation mechanisms were sound throughout; what changed between the two attempts was the amount of real ground truth available to exercise them, not the underlying algorithm. This is a useful distinction for interpreting a system such as this one: a pipeline that appears to produce no effect can be failing for two very different reasons, a broken mechanism or insufficient input data, and only the latter applied here.

The usability study (Section 4.2) points to a separate, genuine limitation of the routing component that the quantitative H3 result does not capture: three of ten participants found aspects of the Safer Route Planner's controls unclear, most specifically the risk-aversion weighting parameter and the on-map pin selection. This is an interface-clarity issue,

not a mechanism issue – the underlying routing logic was independently validated quantitatively – but it is a real barrier to a first-time user and should be addressed before any wider deployment.

5.5 Limitations

Several limitations should be acknowledged directly. The held-out Sofia test set used for H1's domain-adaptation comparison and for H2 contains 42 images, including only 11 pothole instances; per-class Sofia metrics, particularly for the pothole class, should be read as directional rather than statistically precise. H2's ground-truth labels and H4's segment ratings were each assigned by a single rater; a second independent rater, with inter-rater agreement reported, would strengthen both results. RDD2022 itself contains no pothole images from Norway, the United States, or the Czech Republic, so the country-balancing procedure in Section 3.1 could only balance that class across the four countries in which it exists at all, a constraint of the source dataset rather than of the sampling procedure. Faster R-CNN's overfitting pattern (Section 5.1) was not resolved by increasing dataset scale within the range tested in this thesis, and it remains possible that either substantially more data, a different learning-rate schedule, or a smaller backbone would be needed for it to compete with YOLOv8n on this task. H3's exposure measure and the routing cost it evaluates are both computed from the same recorded detection risk (Section 3.8), so the supported result shows that risk-aware routing avoids recorded damage, not that it avoids all road damage in the area; an independent exposure measure, for example from a rater walking each route, would be needed to rule out this circularity. The Sveta Troitsa evaluation boundary used for H3 and H4 was itself derived from the GPS extent of the second photograph set (Section 3.6), so the sampled routes and rated segments lie inside an area shaped by the same data collection being evaluated. Finally, H3 and H4's supported results are geographically scoped to one neighbourhood, chosen for the feasibility of collecting real ground truth within the available time, and are not claimed to generalise to the whole of Sofia without further data collection in additional neighbourhoods.

5.6 Practical implications

For a municipality considering a system of this kind, three practical conclusions follow from these results. First, a lightweight, single-stage detector such as YOLOv8n is the more suitable choice than a two-stage detector such as Faster R-CNN at realistic local training-data volumes, on both accuracy and computational grounds, which matters directly for deployment on modest hardware. Second, a rule-based risk-scoring and routing pipeline can be built and validated end to end without requiring a large, purpose-built local dataset: a domain-adaptation step using around one hundred local photographs already produces a substantial accuracy improvement, and a data-collection effort covering a single neighbourhood was sufficient to make both the routing and risk-heatmap components statistically demonstrable. Third, and most directly actionable, the value of the routing and prioritisation components scales with how much real local detection data has been collected: the system's underlying logic was shown to work correctly as soon as it was given enough real data to exercise it, so the practical bottleneck for a wider rollout is data coverage rather than further algorithmic development.

5.7 Future work

Three directions follow directly from the analysis above, rather than constituting an open-ended list. First, since H3 and H4 were only made testable by concentrating data collection in one neighbourhood, repeating the same photo-collection and processing procedure across additional neighbourhoods is the direct way to test whether the current, positive results generalise across Sofia rather than to one specific area. Second, since H2's shortfall was traced to detection recall rather than the scoring rule (Section 5.3), targeted collection of additional pothole and alligator-crack photographs from the pilot area, the two classes on which recall matters most for the highest-severity risk categories, is a more direct route to improving H2's result than modifying the risk-scoring formula itself. Third, since both H2 and H4 currently rely on a single rater, recruiting a second independent rater for both and reporting inter-rater agreement would directly address the single largest methodological limitation identified in Section 5.5, without requiring any change to the underlying system.

6 Ethical Considerations

6.1 Privacy of individuals and property

Road photographs collected for this thesis may incidentally capture people, faces, vehicles and licence plates, or private property. Both collected sets (Section 3.6) were captured with the same smartphone. The first set was recorded with the camera mounted inside the car, pointed slightly downward and a few metres ahead of the vehicle, which keeps most frames centred on the road surface itself rather than on pedestrians or property. The second set was not captured from a moving vehicle; individual damaged sections were photographed deliberately within the pilot neighbourhood. This capture geometry was not, however, a

strict anonymisation protocol: no explicit step was taken to detect or exclude frames containing people, faces, vehicles, or licence plates, and because the first set was extracted from continuous driving video covering public roads, it is likely that some frames incidentally contain such identifiable content. The current prototype does not implement any automated anonymisation (e.g. face or plate blurring) of stored or displayed images; this is a concrete gap that should be addressed before any deployment beyond a local, single-user prototype, and before any wider release of the raw image sets themselves.

6.2 Responsible data handling

Uploaded images and their associated records are stored locally (a content-hash-derived filename on disk and a row in a local SQLite database, Section 3.5) rather than on a third-party service, which limits external exposure during prototype development. A stored GPS coordinate links a specific photograph to a precise real-world location, which can itself be personally identifying information, for example if it corresponds to a private residence. Any deployment beyond the current local prototype should define an explicit access-control and data-retention policy for this data; no such policy is currently implemented, since it was out of scope for a local, single-user prototype.

Because both manually collected photograph sets (Section 3.6) were captured in Sofia, Bulgaria, an EU member state, the General Data Protection Regulation (GDPR) applies to any personal data they contain. GPS coordinates and photographs that may incidentally show identifiable people or licence plates (Section 6.1) both qualify as personal data under GDPR Article 4(1). The current prototype does not implement a GDPR-compliant workflow: no lawful basis for processing incidentally captured third parties' data has been documented, no privacy notice was provided to anyone photographed, and there is no mechanism for a data subject to request access to or erasure of their data (Articles 15 and 17). These gaps are acceptable for a local, single-user academic prototype evaluated by the author and a small number of usability-study participants, none of whose own images were processed, but any deployment beyond this stage – particularly one operated by a municipality – would need a documented lawful basis (most plausibly *public task*, Article 6(1)(e), for a public authority), a privacy notice, a defined retention schedule, and a working data-subject-request process before processing real residents' images and locations.

6.3 Dataset licensing and attribution

RDD2022 (Arya et al., 2024, 2022b) is used as the primary training dataset under the Creative Commons Attribution 4.0 International licence (CC BY 4.0; `https://creativecommons.org/licenses/by/4.0/`). The manually collected Sofia photographs (Section 3.6) were produced specifically for this thesis and are the author's own data.

6.4 Potential sources of bias

Two sources of bias are already identified and discussed elsewhere in this thesis and are referenced rather than repeated here. First, RDD2022's country-of-origin composition means pothole examples are structurally unavailable from three of the seven source countries (Section 3.1), so trained models' pothole detection is disproportionately shaped by the countries in which pothole examples do exist. Second, the supported H3 and H4 results are geographically scoped to a single Sofia neighbourhood (Section 5.5), so conclusions about routing and risk-heatmap performance should not be assumed to generalise to other areas without further data collection.

6.5 Limitations and risks of automated road-damage assessment

The detection-recall limitation quantified for H2 (Section 4.1.4) means the system can miss real road damage, including high-severity instances. An automated risk score or route recommendation based on an incomplete set of detections carries a specific risk: a road segment with no recorded risk may genuinely be undamaged, or it may simply not yet have been photographed or correctly detected, and the system cannot currently distinguish between these two cases. Outputs should therefore be communicated to users as a best-effort, probabilistic assessment based on the data collected so far, not as a definitive or exhaustive safety judgement.

6.6 The role of human oversight

Given the limitations above, the system described in this thesis is intended to support, rather than replace, official road-safety inspection and maintenance-prioritisation decisions made by qualified municipal personnel. Automated outputs, including risk scores, prioritised segment lists, and route recommendations, should be treated as a decision-support input requiring human review before action, particularly before committing maintenance resources on the basis of a high-risk classification or directing drivers around a specific route.

6.7 Use of generative AI

Generative AI assistance (Claude Sonnet 5; Anthropic 2026) was used during development of the prototype and preparation of this thesis, including code generation, debugging, and drafting and editing of text. All numerical results and statements were checked against the project data and analysis outputs before inclusion.

7 Conclusion

This thesis set out to answer how computer vision-based road damage detection can be integrated with risk-based geospatial routing to improve route safety and support road maintenance prioritisation, through four hypotheses covering detection, risk scoring, routing, and maintenance-prioritisation support.

H1 is partially supported: YOLOv8n did not match Faster R-CNN's accuracy within the predicted two-percentage-point margin, instead outperforming it by a wide, statistically confirmed margin, while also training and running substantially faster, so the predicted efficiency advantage held while the predicted near-parity in accuracy did not. H2, that a rule-based risk score would separate road segments into low, medium, and high risk at a target F1 of 0.75, is not supported at that threshold, but the shortfall was traced to a specific and addressable cause, missed detections rather than mis-scored ones, rather than to a flaw in the scoring rule itself. H3, that risk-aware routing reduces hazard exposure without an unreasonable increase in distance, is supported once real detection coverage was concentrated in one pilot neighbourhood, with a mean exposure reduction of 45.6% and 95% of routes remaining within a 10% distance-increase constraint. H4, that the system enables identification of high-risk segments capable of supporting maintenance prioritisation, is supported, with generated segment risk correlating significantly with independently assessed road condition.

Taken together, these results support a specific, practically grounded conclusion: a lightweight single-stage detector, adapted to its deployment area with a small amount of local data, combined with a simple, explainable risk-scoring and routing pipeline, is both accurate and computationally practical enough to support road maintenance decision-making. The risk-scoring and routing mechanisms themselves were shown to function correctly once given sufficient real ground truth to act on; the principal remaining constraint on the system's usefulness is the volume of real, locally collected detection data, and detection recall specifically, rather than any weakness in the underlying architecture or logic. This connects the four hypotheses into a single finding: the pipeline built in this thesis works as designed, and its next-most-valuable improvement is more real local data, not a different algorithm.

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A Dataset balancing tables

Table A.1 shows the country-of-origin composition of the large training dataset after the fair-share balancing procedure described in Section 3.1.

Table A.1: Large-dataset training images by country of origin, after fair-share class balancing (4 291 images total, including 200 damage-free images not attributed to a class).

Country / source	Images
Japan	810
China (drone)	759
India	723
China (motorbike)	559
Norway	544
United States	477
Czech Republic	419

Class D40 (pothole) draws exclusively from Japan, India, and the two Chinese sources (315, 314, 314, and 157 training images respectively), since RDD2022 contains no pothole images from Norway, the United States, or the Czech Republic; all other classes are balanced across all seven sources.

B Risk-scoring configuration

Table B.2 lists every numeric parameter used by the risk-scoring and routing pipeline (Sections 3.3 and 3.4), all defined in one configuration file so that they can be tuned without a code change.

Table B.2: Risk-scoring and routing configuration parameters.

Parameter	Value
Confidence floor ($m_{\min}$)	0.7
Confidence threshold (τ)	0.25
Damping factor (β)	0.3
Observation-risk ceiling	3.0
Low/medium boundary	0.34
Medium/high boundary	0.67
Routing weight (α)	15 000
Max. segment-matching distance	50 m

C Qualitative usability study – full data and coding

Figure C.1 plots the closed-question means.

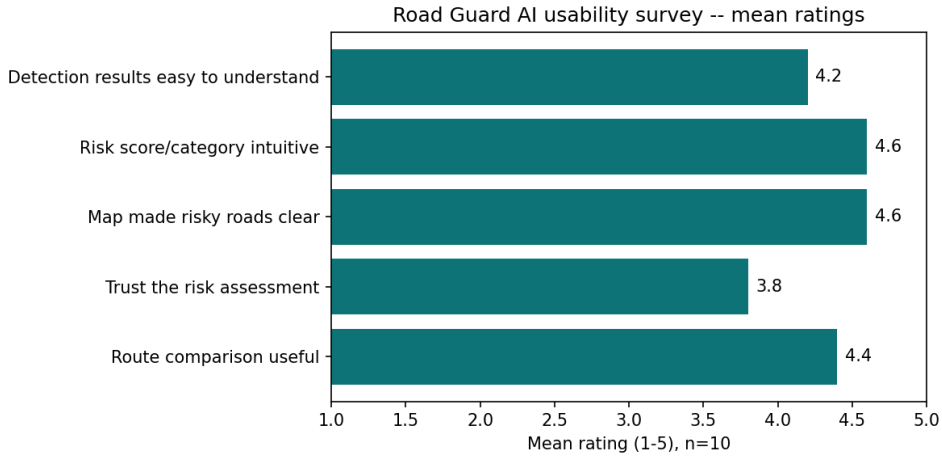


Figure C.1: Mean ratings for the five closed usability questions on a 1–5 scale ($n = 10$). The values match Table 16.

Raw survey responses ($n = 10$) are provided in `thesis/Road_Guard_AI_Usability_Study_Responses_Qualitative_Results.csv`. Descriptive statistics and the chart underlying Table 16 are in `outputs/reports/qualitative_survey_stats.csv` and `outputs/reports/qualitative_survey_means_n10.png`. The full thematic coding, including every supporting quotation for all seven themes summarised in Section 4.2, is provided in `thesis/qualitative_thematic_analysis.md`.

D Application screenshots

The following screenshots were captured from the running Streamlit prototype, using the fine-tuned YOLOv8n model (`yolov8n_large_sofia_finetuned_v1`) and the Sofia observations collected for this thesis.

Figure D.1 and Figure D.2 show the Dashboard page, Figure D.3 the Upload and Detect page, Figure D.4 the Damage Map, and Figure D.5 the Safer Route Planner.

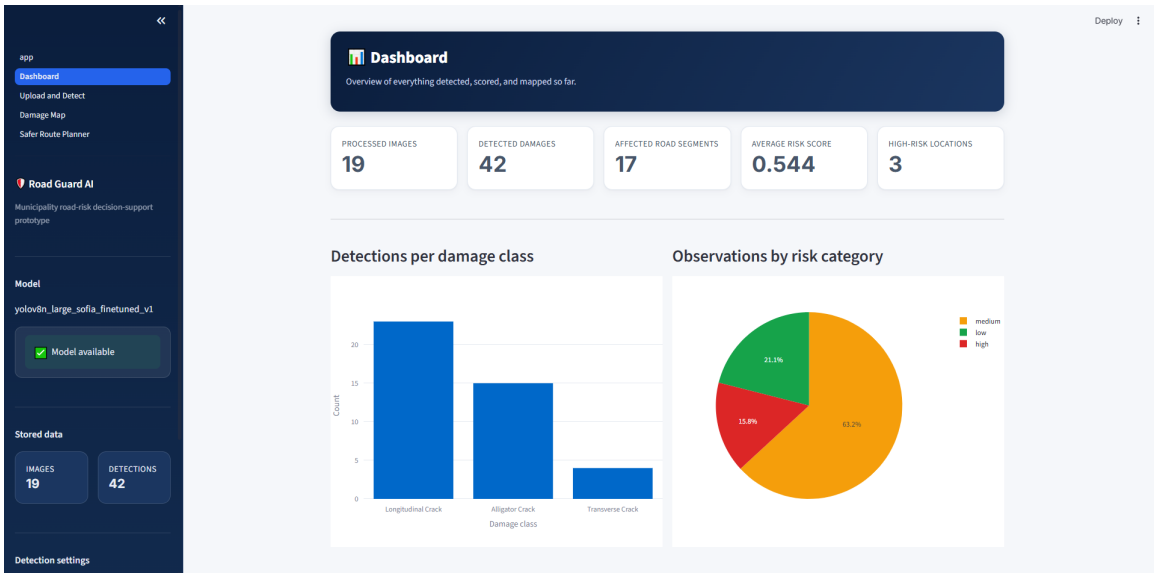


Figure D.1: Dashboard summary: 19 processed images, 42 detected damages, 17 affected road segments, average risk score 0.544, and 3 high-risk locations, with detections per damage class and observations by risk category.

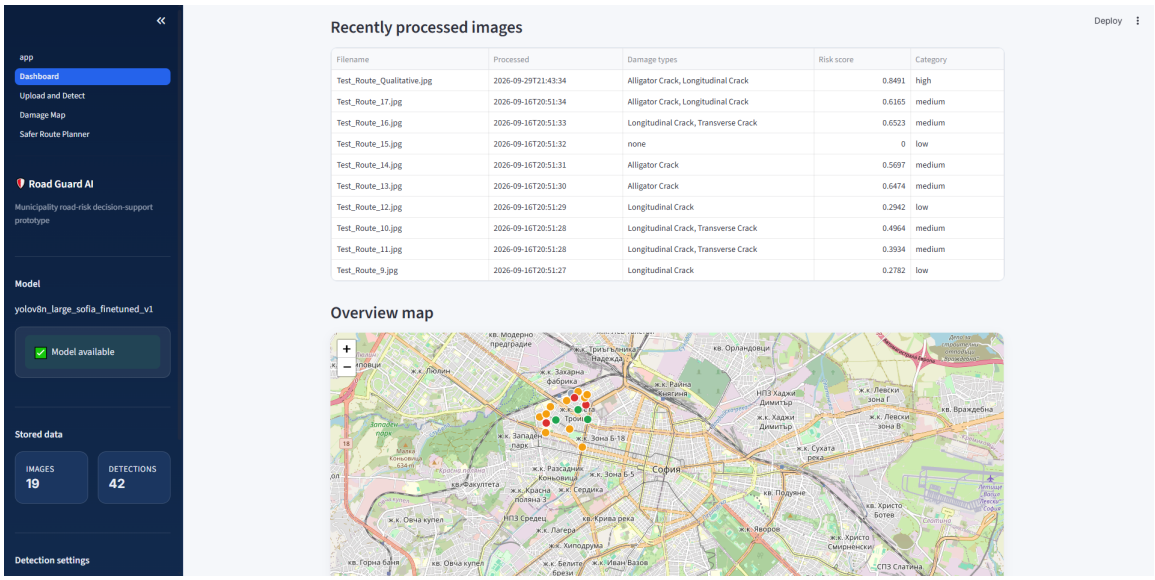


Figure D.2: Dashboard table of recently processed images (filename, damage types, risk score, and category) and overview map of processed observations, clustered around the Sveta Troitsa neighbourhood and colour-coded by risk category.

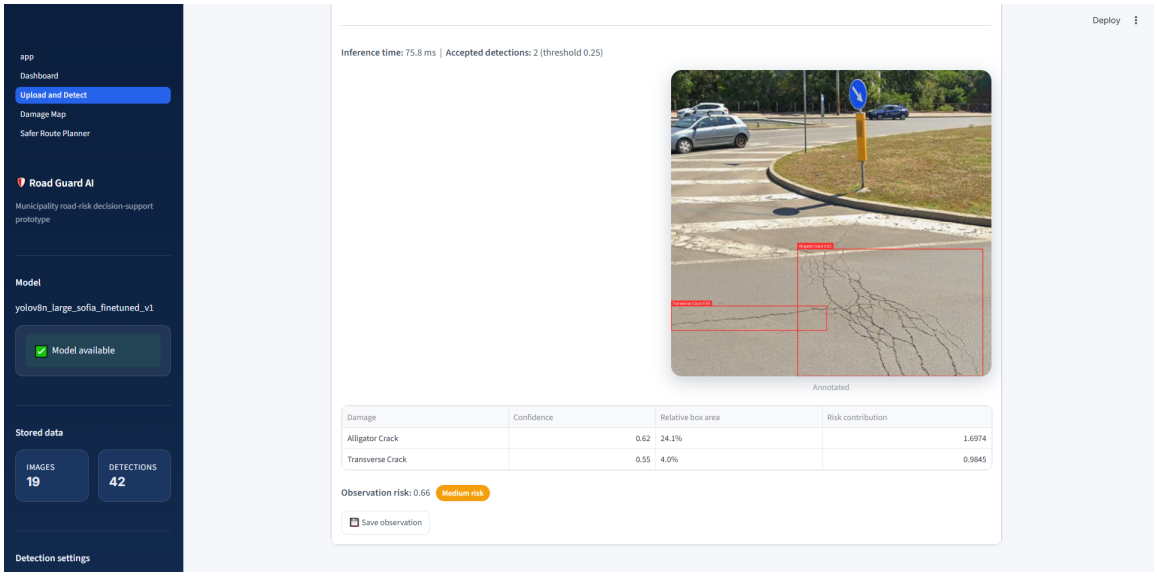


Figure D.3: Upload and Detect page for a single street image: annotated detections with confidence and per-detection risk contribution, and an observation risk of 0.66 (medium).

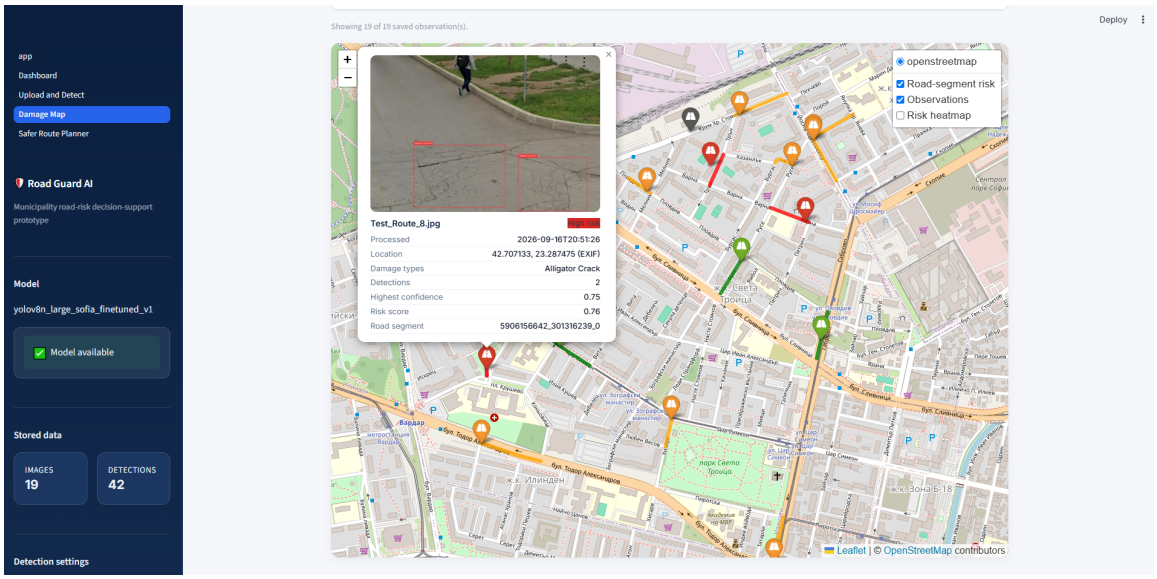


Figure D.4: Damage Map with the road-segment risk layer and observation markers in the Sveta Troitsa neighbourhood. The open popup shows one observation's EXIF location, detections, and segment risk (0.76, high).

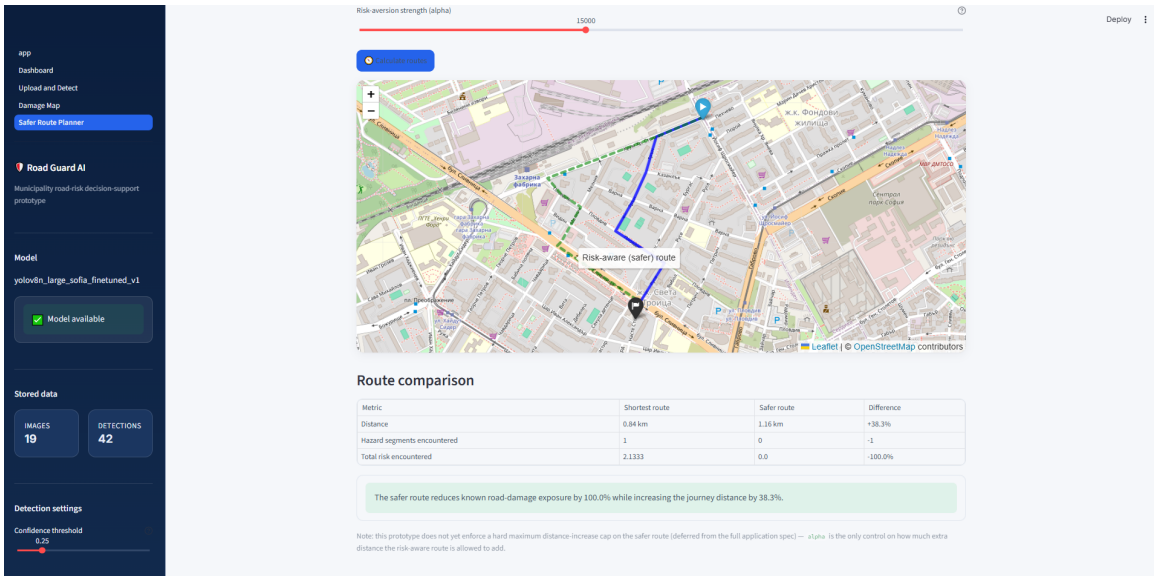


Figure D.5: Safer Route Planner comparing the shortest (blue) and risk-aware (green) route for one origin–destination pair in the Sveta Troitsa neighbourhood (risk-aversion strength $\alpha = 15\,000$). This is a single illustrative example, not a representative route; aggregate routing results are reported in Section 4.1.5.